

Math Review

ECON 3070 - Intermediate Microeconomic Theory

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Math Review Topics

- Limits
- Derivative Definition
- Derivative Rules
- Partial Derivatives

The limit of $f(x)$ as x approaches a is written as:

$$\lim_{x \rightarrow a} f(x)$$

- The behavior of the function as its input approaches a
- In certain cases, can be evaluated by plugging in $x = a$
- $\lim_{x \rightarrow a} c = c$ for a constant c

Limits

The limit from the left and right are

→ $\lim_{x \rightarrow a^-} f(x) = A$ "as x approaches a from the left ($-\infty$)"

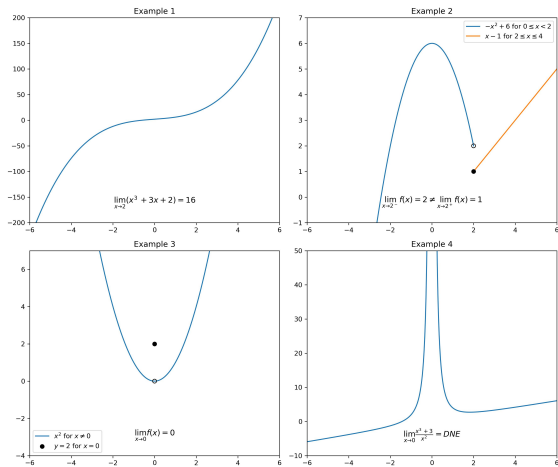
→ $\lim_{x \rightarrow a^+} f(x) = A$ "as x approaches a from the right (∞)"

If $\lim_{x \rightarrow a^-} f(x) = A = \lim_{x \rightarrow a^+} f(x)$ then the limit exists and we write

$$\lim_{x \rightarrow a} f(x) = A$$

- Note: the limit does not exist if $A = \pm\infty$, even if the left and right limit both equal $\pm\infty$

Limits



$$\lim_{x \rightarrow a} f(x) = L \quad \lim_{x \rightarrow a} g(x) = M$$

- **Constant Multiple Rule:** $\lim_{x \rightarrow a} [af(x)] = aL$
- **Sum/Difference Rule:** $\lim_{x \rightarrow a} [f(x) \pm g(x)] = L \pm M$
- **Product Rule:** $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = L \cdot M$
- **Quotient Rule:** $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{L}{M}, \quad M \neq 0$
- **Power Rule:** $\lim_{x \rightarrow a} [(f(x))^n] = L^n, \quad n > 0$

Definition of a Derivative

The derivative of f is defined as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

If this limit exists, then we say f is differentiable at x .

Derivative Notation

All of these mean “take the derivative of f with respect to x ”:

→ $f'(x)$

→ $\frac{df(x)}{dx}$

→ $\frac{d}{dx}f(x)$

→ $\frac{dy}{dx}$ if $y = f(x)$

→ y' (sometimes)

→ $\dot{y}$ (particularly in Macro)

Derivative Rules

Let $f(x)$ and $g(x)$ be differentiable functions:

- **Derivative of a Constant:** $f(x) = c, f'(x) = 0$
- **"Power Rule":** $f(x) = x^n, f'(x) = nx^{n-1}$
- **Sum/Difference Rule:** $\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$
- **Product Rule:** $\frac{d}{dx}[f(x) \cdot g(x)] = f'(x) \cdot g(x) + f(x) \cdot g'(x)$
- **Quotient Rule:** $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$
- **Chain Rule:** $\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$

Derivatives Practice Problems

$$f(x) = 2x$$

Derivatives Practice Problems – Solution

$$f(x) = 2x$$

By the power rule ($f(x) = 2x^1$, so $n = 1$):

$$f'(x) = 2$$

Derivatives Practice Problems

$$f(x) = 3x^2$$

Derivatives Practice Problems – Solution

$$f(x) = 3x^2$$

By the power rule ($n = 2$):

$$f'(x) = 3 \cdot 2x^{2-1} = 6x$$

Derivative Rules for Exponents and Natural Log

$$\rightarrow f(x) = e^x, f'(x) = e^x$$

$$\rightarrow f(x) = \ln(x), f'(x) = \frac{1}{x}$$

Useful review of the properties of **exponents** and **logs**.

Derivatives Practice Problems

$$f(x) = \ln(e^x)$$

Derivatives Practice Problems – Solution

$$f(x) = \ln(e^x)$$

Simplify first, using the log property $\ln(e^x) = x$:

$$f(x) = x \implies f'(x) = 1$$

(Or, via the chain rule directly: $f'(x) = \frac{1}{e^x} \cdot e^x = 1$ – same answer either way.)

Derivatives Practice Problems

$$f(x) = x^2 \ln(x)$$

Derivatives Practice Problems – Solution

$$f(x) = x^2 \ln(x)$$

Use the product rule, with $u = x^2$ ($u' = 2x$) and $v = \ln(x)$ ($v' = 1/x$):

$$f'(x) = 2x \ln(x) + x^2 \cdot \frac{1}{x} = 2x \ln(x) + x$$

Derivatives Practice Problems

$$f(x) = e^{2x}$$

Derivatives Practice Problems – Solution

$$f(x) = e^{2x}$$

Use the chain rule, with outer function e^u and inner function $u = 2x$ ($u' = 2$):

$$f'(x) = e^{2x} \cdot 2 = 2e^{2x}$$

Higher-order Derivatives

The second-order derivative is

$$f''(x) = \frac{d}{dx} f'(x)$$

We could keep differentiating as long as the last derivative is differentiable. Notation:

- $f''(x)$
- $\frac{d^2 f(x)}{dx^2}$
- $\frac{d^2}{dx^2} f(x)$
- $\frac{d^2 y}{dx^2}$ if $y = f(x)$
- y'' (sometimes)

Partial Derivatives

The function $z = f(x, y)$ takes two inputs and outputs a third. We can evaluate what happens to that output when we change x or y (but not both).

- $\frac{\partial z}{\partial x} =$ “the derivative of z with respect to x ”
 - The slope of f in the cardinal direction of x (e.g. “north-south”)
 - The tangent of f in the x direction at a point (x_0, y_0)
- $\frac{\partial z}{\partial y} =$ “the derivative of z with respect to y ”
 - The slope of f in the cardinal direction of y (e.g. “east-west”)
 - The tangent of f in the y direction at a point (x_0, y_0)

Partial Derivatives

All of these mean “take the partial derivative of f with respect to x ”:

- $f_x(x, y)$
- $\frac{\partial f(x, y)}{\partial x}$
- $\frac{\partial}{\partial x} f(x, y)$
- $\frac{\partial z}{\partial x}$ if $z = f(x, y)$

Partial Derivatives

Interpret $\frac{\partial z}{\partial x}$ as “what happens to z if I change x , holding y constant.” Likewise for $\frac{\partial z}{\partial y}$.
Suppose $z = x + y + xy$, find $\frac{\partial z}{\partial x}$.

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}[x + y + xy] = \frac{\partial}{\partial x}x + \frac{\partial}{\partial x}y + \frac{\partial}{\partial x}xy = 1 + 0 + y \cdot 1 = 1 + y$$

Partial Derivatives Practice Problems

$$f(x, y) = x^2 + y^2 + x^2y^2$$

Partial Derivatives Practice Problems – Solution

$$f(x, y) = x^2 + y^2 + x^2y^2$$

Holding y constant, differentiate term-by-term with respect to x :

$$\frac{\partial f}{\partial x} = 2x + 0 + 2xy^2 = 2x(1 + y^2)$$

Holding x constant, differentiate term-by-term with respect to y :

$$\frac{\partial f}{\partial y} = 0 + 2y + 2x^2y = 2y(1 + x^2)$$

Partial Derivatives Practice Problems

$$f(x, y) = x^2y + \ln(x)y^3$$

Partial Derivatives Practice Problems – Solution

$$f(x, y) = x^2y + \ln(x)y^3$$

Holding y constant, differentiate with respect to x (treating y, y^3 as constants):

$$\frac{\partial f}{\partial x} = 2xy + \frac{1}{x}y^3$$

Holding x constant, differentiate with respect to y (treating $x^2, \ln(x)$ as constants):

$$\frac{\partial f}{\partial y} = x^2 + 3\ln(x)y^2$$

Properties of Exponents

Let a and b be real numbers and m and n be integers. Then the following properties of exponents hold, provided that all of the expressions appearing in a particular equation are defined:

$$\rightarrow a^m a^n = a^{m+n}$$

$$\rightarrow (a^m)^n = a^{mn}$$

$$\rightarrow (ab)^m = a^m b^m$$

$$\rightarrow \frac{a^m}{a^n} = a^{m-n}, a \neq 0$$

$$\rightarrow \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}, b \neq 0$$

$$\rightarrow a^{-m} = \frac{1}{a^m}, a \neq 0$$

$$\rightarrow a^{\frac{1}{n}} = \sqrt[n]{a}$$

$$\rightarrow a^0 = 1, a \neq 0$$

$$\rightarrow a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

Properties of Natural Log

Note that logs are only defined for positive values of x :

$$\rightarrow \ln(xy) = \ln(x) + \ln(y)$$

$$\rightarrow \ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$$

$$\rightarrow \ln(x^y) = y \cdot \ln(x)$$

$$\rightarrow \ln(e^x) = x$$

$$\rightarrow e^{\ln(x)} = x$$

$$\rightarrow \ln(e) = 1$$

$$\rightarrow \ln(1) = 0$$

$$\rightarrow \ln(0) = \textit{undefined}$$